A concentration inequality for suprema of conditional Gaussian processes*

Victor Amaya† Haotian Gu‡

Abstract

new: We ask how far from observed data Gaussian process regression remains informative, and we answer this non-asymptotically for noiseless regression under a stationary prior. A centred prior $f \sim \mathcal{GP}(0, \phi)$ is conditioned on n exact observations recorded at fixed design points, producing a non-stationary posterior $g \sim \mathcal{GP}(\mu_n, s_n^2)$ that interpolates the data. Far from the design points, the posterior should revert to the prior, and our main result makes this reversion quantitative. We compare the two processes through their suprema, the natural object of the extreme value theory of Gaussian processes, and the functional to which Gaussian concentration of measure, via the Borell–TIS inequality, applies. Coupling f and g on a common probability space and writing M_A^f and M_A^g for their suprema over a compact set A at distance δ from the design, we prove a non-asymptotic, sub-Gaussian tail bound for $|M_A^f - M_A^g|$ at the scale $\phi(\delta)$. Inverting it yields an explicit *certainty distance*, i.e., a computable radius beyond which the prior and posterior suprema agree to any prescribed accuracy and confidence, certifying where the regression stops being informative about extreme values. It is a certificate of uninformativeness, not of accuracy. In general the decay rate depends on the number of observations n; when $\log \phi$ is concave the constants in the decay rate become independent of n.

Keywords: Gaussian process; supremum of a Gaussian process; concentration of measure; Gaussian process regression; non-stationary covariance; Borell–TIS inequality; certainty distance.
MSC2020 subject classifications: 60G15; 60G70; 60E15; 62M20.

1 Introduction

new: Gaussian process regression (GPR) occupies a central place in contemporary statistics. It is the kriging predictor of spatial statistics [6], the smoothing spline of penalized least squares [20], the surrogate that drives Bayesian global optimization [19],

*Dedicated to the memory of Sayan Mukherjee

†Duke University. E-mail: vamayacar@gmail.com
‡University of California, Los Angeles. E-mail: htg@math.ucla.edu

and a canonical nonparametric Bayes procedure [16]. Its appeal is that it returns not a single point prediction but a full conditional distribution, and with it a quantification of predictive uncertainty at every input. This quantification comes with a caveat, that is well known in practice and visible in Figure 1, predictions are trustworthy only near the observed inputs and revert to the prior far from them. Practitioners use it as a warning against extrapolation, and a large computational literature built on Vecchia approximations and nearest-neighbor Gaussian processes [9, 5] rests on the implicit premise that sufficiently distant observations can be discarded. In both, this distance is treated as a heuristic rather than a quantity one can explicitly compute. Which rises the natural question: *At what distance from the data does the regression cease to add information to the prior?* This paper supplies such quantity, in closed form for the Gaussian kernel, together with a probabilistic guarantee.

We work in the standard noiseless Bayesian setting, which isolates the geometric phenomenon at issue. A stationary Gaussian prior $f \sim \mathcal{GP}(0, \phi)$, conditioned on n exact observations $X_{1:n}$ at fixed design points $\{t_i\}$, yields a *non-stationary* posterior $g \sim \mathcal{GP}(\mu_n, s_n^2)$ that interpolates the data; the formal construction and the update formulas are recalled in Section 2. The relevant geometric quantity is the distance $\delta = \mathrm{dist}(A, \{t_i\}_{i=1}^n)$ from a compact region A to the observed inputs, and all probabilities are conditional on $X_{1:n}$.

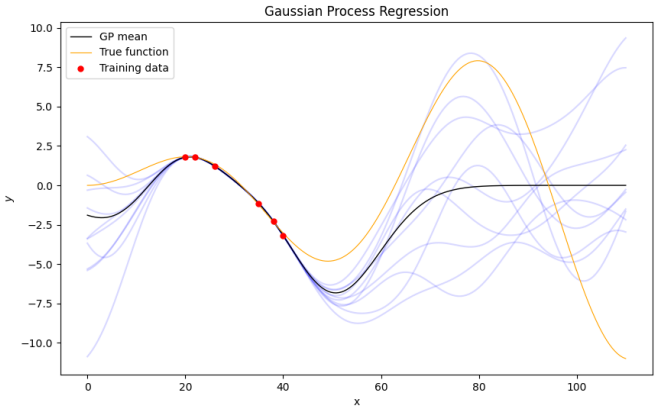


Figure 1: The orange curve represents the data-generating function and the red dots are sampled noiseless observations. Posterior sample paths (light blue) coincide near the data and fan out as the distance to the observations grows; the posterior mean (black) reverts to the prior mean far from the data.

To Make the question precise a comparison criterion is required, and ours is the *comparison of suprema*, motivated by extreme value theory. In the settings where extrapolation matters most, among them environmental monitoring, structural reliability, and safety-critical control, the quantity of interest over an unexplored region is an extreme, not a typical value. Namely, what one wants to approximate is the largest value the unknown function may take there, which is the natural object of extreme value theory [17, 18]. For Gaussian processes the supremum is moreover one of most studied extremal functional [1]. For a compact A disjoint from the design we compare $M_A^f = \sup_{t\in A} f(t)$ with $M_A^g = \sup_{t\in A} g(t)$, placing the two draws on a common probability space by the coupling of Lemma 3.6, so that $|M_A^f - M_A^g|$ is a genuine random variable and the comparison is one of joint realizations rather than of laws. The general intuitive idea is the following: the closer the two suprema, the less the observations tell us about

the extreme behavior of the unknown function on A.

1.1 Main results

The contributions are fourfold. First, we prove a non-asymptotic concentration inequality for $|M_A^f - M_A^g|$, sub-Gaussian at the scale $\phi(\delta)$ (Theorem 2.6). Second, inverting this inequality yields the *certainty distance*, an explicit radius beyond which the regression is certifiably uninformative about extremes at any prescribed accuracy τ and confidence $1-\varepsilon$ (Section 4). Third, under log-concavity of ϕ we sharpen the bound to one in which the constant in the exponent is independent of n and asymptotically optimal as the design sparsifies (Theorem 5.9, Corollary 5.10, Remark 5.11), and we give a dimension-explicit generalized version on $\mathbb{R}^d$ (Theorem 5.13) for any finite dimension of the index set. Fourth, as a byproduct, we obtain extremal concentration estimates with constants uniform in n for the non-stationary processes produced by conditioning. To our knowledge these are new, since the sharp extremal results closest to ours concern stationary fields. As a byproduct, we obtain extremal concentration estimates with constants uniform in n for the non-stationary processes produced by conditioning.

Theorem (General bound)**.** Under the standing assumptions of Section 2, with $\{t_i\}_{i=1}^n$ ρ-separated and non-degenerate, there are constants $C, c > 0$ depending only on n, ρ such that for every $\tau > 0$ and every sufficiently large δ,

$$\mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \leq C \exp\left(-\frac{c\,\tau^2}{\phi(\delta)^2}\right). \tag{1.1}$$

The precise statement is Theorem 2.6.

Since ϕ is strictly decreasing, the right-hand side tends to 0 as $\delta \to \infty$. Far from the data the conditional process is, in terms of its supremum, indistinguishable from the unconditional one, and (1.1) may be solved for δ to give the certainty distance (Section 4). The certificate is deliberately one-sided. It locates where regression adds nothing to the prior, and says nothing about how well it predicts closer in.

The constant c in (1.1) depends on n, both directly and through a non-degeneracy condition that forces the separation ρ to grow with n. As it is, this would be troubling, since it would make the certainty distance grow with the sample size, as if remote observations extended the model's reach. Our second result shows the dependence to be an artifact of the proof. Because ρ-separation forces the cross-covariances to decay geometrically, the crude counting estimates behind (1.1) can be replaced by summable lattice sums. Theorem 5.9 then shows that, when $\log \phi$ is concave, the bound holds with a constant depending only on ρ, so that additional well-separated data fill in the interior of the influence region without pushing out its boundary. The constant is asymptotically optimal, approaching $1/4$ in the sparse-design limit, where a two-point design shows that no uniform bound can do better (Remark 5.11). We caution that n-freedom is a statement about the *rate* and not the hypothesis. The distance at which the bound activates still depends on the sample through $\|X_{1:n}\|_\infty$, and on n through a $\sqrt{\log n}$ factor that is shown to be unavoidable (Remark 5.8).

1.2 Related work

For a single *stationary* field, Tanguy [3] sharpened Gaussian concentration into tail inequalities reflecting the true fluctuation rate of the maximum, via superconcentration [10]. Anti-concentration inequalities for Gaussian maxima [11, 12] quantify the concentration function of the maximum and thereby convert such comparisons into bounds in Kolmogorov distance. Our problem differs in object and method, since what we do is compare a stationary prior with its non-stationary conditional version, coupled on the

same probability space, and bound the discrepancy of their suprema. On the regression side, Lederer et al. [13, 14] bound the posterior error and variance using nearby data to certify where GPR is *accurate*. We instead quantify, dually, where it is *uninformative*, by comparing extrema over a region rather than a pointwise summary. Liu and Kok [15] bound expected extremes over unseen inputs but neither compare prior with posterior nor a cutoff distance, in terms of the models parameters. The posterior-contraction literature [16] treats the opposite, data-rich regime. Classically, the locality of conditioning is captured by Stein's screening effect [7, 8] and the effective range [6]. Both of these results are qualitative or heuristic, and it is presupposed by the scalable near-neighbor approximations [5]. Section 6 relates our certainty distance to them. To our knowledge a non-asymptotic concentration inequality for the discrepancy between the suprema of a Gaussian prior and its posterior, and the certainty distance it yields, were not studied before.

2 Setup and assumptions

Let $(T, \|\cdot\|)$ be a compact subset of $\mathbb{R}^d$, which serves as the index set of the process (a set of times or spatial locations). Let $\phi\colon T \times T \to \mathbb{R}$ be the prior covariance function, assumed stationary (isotropic). That is, there is a radial profile, still denoted ϕ, with $\phi(s,t) = \phi(\|s-t\|)$. We write $\phi(r)$ for the profile evaluated at a distance $r \geq 0$. Isotropy holds for the Matérn and Gaussian families that dominate applications, and it is what lets every bound below be stated as a function of the single scalar distance δ rather than of the relative geometry of A and $\{t_i\}$.

The data are the noiseless values $X_{1:n} = (X_{t_1}, \ldots, X_{t_n})$ recorded at a fixed, non-random collection of ρ-separated inputs $\{t_i\}_{i=1}^n \subseteq T$ satisfying (D_n) (Definition 2.3); they are produced by an unknown mechanism and are treated as given. We model the unknown function behind them by placing on the functions on T the Gaussian prior distribution $f \sim \mathcal{GP}(0, \phi)$, with prior mean function 0 and prior covariance function ϕ. Conditioning this prior on the data $X_{1:n}$ produces the posterior distribution $g \sim \mathcal{GP}(\mu_n, s_n^2)$, with

$$\begin{aligned} \mu_n(t) &= \phi(t_{1:n}, t)^\top \, \phi(t_{1:n}, t_{1:n})^{-1} \, X_{1:n}, \\ s_n^2(s,t) &= \phi(s,t) - \phi(t_{1:n}, s)^\top \, \phi(t_{1:n}, t_{1:n})^{-1} \, \phi(t_{1:n}, t), \end{aligned} \tag{2.1}$$

where the cross-covariance vector is $\phi(t_{1:n}, t) = (\phi(t_1, t), \ldots, \phi(t_n, t))^\top \in \mathbb{R}^n$ and the Gram matrix is given by $\phi(t_{1:n}, t_{1:n}) = (\phi(t_i, t_j))_{i,j=1}^n \in \mathbb{R}^{n \times n}$. Throughout, $\|\cdot\|_2$ is the Euclidean norm and $\|\cdot\|_{\mathrm{op}}$ the operator norm. Both formulas in (2.1) are governed by the same object, the inverse Gram matrix $\phi(t_{1:n}, t_{1:n})^{-1}$, and controlling it is the main technical obstacle of the paper; a quantitative lower bound on the smallest eigenvalue of $\phi(t_{1:n}, t_{1:n})$, given in Lemma 3.1 below, is what makes this possible.

Definition 2.1 (Supremum on a set)**.** For a continuous Gaussian process f on T and a compact set $A \subseteq T$, the *supremum of f on A* is the random variable

$$M_A^f \coloneqq \sup_{t \in A} f(t).$$

Definition 2.2 (ρ-separation)**.** The inputs $\{t_i\}_{i=1}^n$ are *ρ-separated* if $\min_{i \neq j} \|t_i - t_j\| \geq \rho$.

We now define the *non-degeneracy condition* on the separation parameter, whose sole role is to keep the Gram matrix $\phi(t_{1:n}, t_{1:n})$ invertible.

Definition 2.3 (Non-degeneracy condition)**.** A ρ-separated design $\{t_i\}_{i=1}^n$ satisfies the *non-degeneracy condition* if

$$(\mathrm{D}_n) \qquad \rho > \phi^{-1}(1/n) > 0.$$

This is assumed in force unless stated otherwise. It is a requirement on how large the separation parameter ρ must be, not part of the notion of ρ-separation itself (Definition 2.2). A weaker, n-free variant, *non-degeneracy condition* is defined in Section 5, stated there once the quantities it involves are available (condition (D), Definition 5.1).

Definition 2.4 (Typical sample)**.** The observed data vector $X_{1:n}$ is called a *typical sample* if $\|X_{1:n}\|_2 \le C_1\sqrt{n}$ for a fixed constant $C_1 > 0$.

Definition 2.4 is a deterministic condition on the recorded data, and it excludes only samples that are atypical *under the model*. That is, if the data are generated by the mechanism the prior describes, equivalently, under the prior predictive law $X_{1:n} \sim \mathcal{N}\big(0, \phi(t_{1:n}, t_{1:n})\big)$, so that $X_{t_i} \sim \mathcal{N}(0, \phi(0))$ with $\phi(0) = 1$, the event in Definition 2.4 carries probability at least $1 - e^{-c'n}$ for a suitable $c' > 0$. Indeed, using Gaussian concentration of the norm, write $X_{1:n} = \phi(t_{1:n}, t_{1:n})^{1/2}\xi$ with $\xi \sim \mathcal{N}(0, I_n)$, the map $\xi \mapsto \|X_{1:n}\|_2$ is Lipschitz with constant $\sqrt{\lambda_{\max}(\phi(t_{1:n}, t_{1:n}))} \le \sqrt{1 + n\phi(\rho)} < \sqrt{2}$ under (D_n), and using as well that $\mathbb{E}\|X_{1:n}\|_2 \le \sqrt{n}$, then any $C_1 > 1$ works by simply taking the constant $c' = (C_1 - 1)^2/4$. Thus, conditioning on a typical sample loses only an exponentially small probability.

Standing assumptions on the kernel. We assume that ϕ satisfies the following.

(A1) *Normalization and positivity:* $\phi(0) = 1$ and $\phi(r) \ge 0$ for all $r \ge 0$.

(A2) *Monotonicity:* $r \mapsto \phi(r)$ is continuous and strictly decreasing, with $\phi(r) \to 0$ as $r \to \infty$; in particular ϕ^{-1} is well defined on $(0, 1]$.

(A3) *Regularity:* ϕ is Lipschitz, $|\phi(r) - \phi(r')| \le L\,|r - r'|$.

None of the three is restrictive in practice, (A1) is a costless normalization of scale, (A2) is what makes *distance to the data* a meaningful quantity, since it is what lets ϕ be inverted into the certainty distance of Section 4, and Assumption (A3) imposes the minimal regularity necessary for the supremum over A to be a well-defined random variable, bypassing technical measurability issues by ensuring the Gaussian process has almost surely continuous sample paths. The Gaussian (radial basis function) kernel $\phi(r) = e^{-r^2/2}$ satisfies (A1)–(A3) and is our running example.

Remark 2.5. An upper bound for the left-hand side of (1.1), once fixed at a prescribed level, plays the role of an *uncertainty threshold*; solving for δ then yields a candidate *certainty distance*. This is made precise in Section 4.

The main result bounds the conditional probability that the suprema of a prior and a posterior draw differ by more than τ. The rate is sub-Gaussian with scale $\phi(\delta)$, the covariance at the distance from A to the data, so the bound sharpens as A recedes from the observed inputs.

Theorem 2.6 (Main result, general form)**.** Let $X_{1:n}$ be a typical sample (Definition 2.4) recorded at ρ-separated inputs $\{t_i\}_{i=1}^n$ satisfying (D_n) (Definition 2.3), with ϕ satisfying (A1)–(A3). Let $f \sim \mathcal{GP}(0, \phi)$ be a draw from the prior and let $g \sim \mathcal{GP}(\mu_n, s_n^2)$ be a draw from the posterior obtained by conditioning that prior on $X_{1:n}$, with mean and covariance as in (2.1), the two being coupled as in Lemma 3.6. Then for every compact $A \subset T$ with $A \cap \{t_i\}_{i=1}^n = \emptyset$, every $\tau > 0$, and every sufficiently large $\delta = \mathrm{dist}(A, \{t_i\}_{i=1}^n)$,

$$\mathbb{P}\big(|M_A^f - M_A^g| \ge \tau \mid X_{1:n}\big) \le 2\exp\left(-\frac{c_{n,\rho}\,\tau^2}{\phi(\delta)^2}\right), \qquad a.s. \tag{2.2}$$

where $c_{n,\rho} > 0$ depends only on ϕ, n and ρ.

The proof of Theorem 2.6 rests on a coupling that places the prior and the posterior on a common probability space, after which the discrepancy $|M_A^f - M_A^g|$ is controlled.

3 Auxiliary results and proof of the n-dependent bound

The proof of Theorem 2.6 requires some quantitative estimates, and we establish them in this section. We start with a lower bound on the smallest eigenvalue of the Gram matrix (Section 3.1), which is what makes the inverse $\phi(t_{1:n}, t_{1:n})^{-1}$ tractable. We then provide bounds on the variance and on the canonical metric of the auxiliary process Z that carries the discrepancy (Section 3.2); and the resulting control of the expected supremum of Z (Section 3.3). The coupling and the proof itself occupy Section 3.4.

3.1 A lower bound for the Gram matrix

We first bound the smallest eigenvalue of the Gram matrix away from zero. Write $\lambda_{\min}(M)$ for the smallest eigenvalue of a symmetric matrix M.

Lemma 3.1 (Lower bound for the Gram matrix)**.** Under (A1)–(A2), let $\{t_i\}_{i=1}^n$ be ρ-separated and let (D_n) hold, i.e. $\rho > \phi^{-1}(1/n)$. Then

$$\lambda_{\min}\big(\phi(t_{1:n}, t_{1:n})\big) \;\geq\; c_0 \coloneqq 1 - n\,\phi(\rho) \;>\; 0,$$

for a c_0 that depends only on ϕ, n and ρ.

Proof. Since $\rho > \phi^{-1}(1/n)$ and ϕ is strictly decreasing, $\phi(\rho) < 1/n$. Using monotonicity and $\phi \geq 0$,

$$\max_{1\leq i\leq n} \sum_{j\neq i} \phi(t_i, t_j) \leq n\,\phi(\rho) < 1. \tag{3.1}$$

The Gram matrix has unit diagonal by (A1). By the Gershgorin-type bound of Johnson [2, Theorem 3] (which for a symmetric positive semidefinite matrix with unit diagonal reduces to Gershgorin's theorem),

$$\lambda_{\min}\big(\phi(t_{1:n}, t_{1:n})\big) \geq 1 - \max_{1\leq i\leq n} \sum_{j\neq i} \phi(t_i, t_j) \geq 1 - n\,\phi(\rho) = c_0 > 0,$$

where positivity is a consequence of (3.1). □

Example 3.2. For the Gaussian kernel $\phi(r) = e^{-r^2/2}$ one has $\phi^{-1}(1/n) = \sqrt{2\log n}$, which grows only logarithmically in the number of observations (even for $n = 10^6$ it suffices to take $\rho > 5.26$, indeed, $\sqrt{2\log 10^6} \approx 5.257$).

3.2 The auxiliary process: variance and canonical metric

We now introduce a centered Gaussian process Z that will be used below, in Lemma 3.6, for the coupling of the prior and posterior processes. Define the covariance of the centered process Z as follows,

$$\psi_Z(x, y) \coloneqq \phi(t_{1:n}, x)^\top\, \phi(t_{1:n}, t_{1:n})^{-1}\, \phi(t_{1:n}, y), \qquad x, y \in T. \tag{3.2}$$

Using (2.1) we can write $\psi_Z = \phi - s_n^2$, so ψ_Z is a genuine covariance kernel. The next lemma controls its size and the associated canonical (pseudo) metric $d_Z(x, y) \coloneqq \big(\mathbb{E}[(Z(x) - Z(y))^2]\big)^{1/2}$ on a set A that is far from the observed inputs.

Lemma 3.3 (Variance and canonical-metric bounds)**.** Let Z be the centered Gaussian process with covariance (3.2), let $A \subseteq T$ satisfy $A \cap \{t_i\}_i = \emptyset$, and set $\delta = \operatorname{dist}(A, \{t_i\}_{i=1}^n)$. Under (A1)–(A3) and the hypotheses of Lemma 3.1, for all $x, y \in A$:

$$\psi_Z(x, x) \leq \tfrac{1}{c_0}\, n\,\phi(\delta)^2, \tag{3.3}$$

$$d_Z(x, y) \leq \tfrac{1}{\sqrt{c_0}}\, \|\, \phi(t_{1:n}, x) - \phi(t_{1:n}, y)\, \|_2 \;\leq \tfrac{L}{\sqrt{c_0}}\, \sqrt{n}\, \|x - y\|. \tag{3.4}$$

In particular $\sigma_Z^2 \coloneqq \sup_{x\in A} \operatorname{Var}\big(Z(x)\big) \le \frac{n}{c_0}\,\phi(\delta)^2$, and the diameter of A in the canonical metric satisfies

$$\operatorname{diam}(A, d_Z) \;\le \frac{2\sqrt{n}}{\sqrt{c_0}}\,\phi(\delta). \tag{3.5}$$

Proof. Write $a(t) \coloneqq \phi(t_{1:n}, t)$ and $M \coloneqq \phi(t_{1:n}, t_{1:n})$. For $t \in A$ and each i we have $\|t_i - t\| \ge \delta$, so by (A1)–(A2) the i-th entry of the vector $a(t)$ satisfies $0 \le a_i(t) = \phi(\|t_i - t\|) \le \phi(\delta)$, whence

$$\|a(t)\|_2 \le \sqrt{n}\,\phi(\delta), \qquad t \in A. \tag{3.6}$$

By Lemma 3.1, $\|M^{-1}\|_{\mathrm{op}} = 1/\lambda_{\min}(M) \le 1/c_0$. The variance bound (3.3) follows from

$$\psi_Z(x,x) = a(x)^\top M^{-1} a(x) \le \|M^{-1}\|_{\mathrm{op}}\,\|a(x)\|_2^2 \le \tfrac{n}{c_0}\,\phi(\delta)^2.$$

For the canonical metric, write $Z(x) = a(x)^\top M^{-1/2}\xi$ with $\xi \sim \mathcal{N}(0, I_n)$, which has the right covariance because $a(x)^\top M^{-1/2} M^{-1/2} a(y) = a(x)^\top M^{-1} a(y) = \psi_Z(x,y)$. Hence

$$d_Z(x,y)^2 = \mathbb{E}\big[(Z(x) - Z(y))^2\big] = \big(a(x) - a(y)\big)^\top M^{-1}\big(a(x) - a(y)\big) \le c_0^{-1}\,\|a(x) - a(y)\|_2^2,$$

which is the first inequality in (3.4). By the reverse triangle inequality $\big|\,\|t_i - x\| - \|t_i - y\|\,\big| \le \|x - y\|$ and (A3),

$$\|a(x) - a(y)\|_2^2 = \sum_{i=1}^{n} \big(\phi(\|t_i - x\|) - \phi(\|t_i - y\|)\big)^2 \le nL^2 \|x - y\|^2,$$

giving the second inequality in (3.4).

Finally, for the diameter, by Cauchy–Schwarz for the covariance,

$$|\psi_Z(x,y)| \le \sqrt{\psi_Z(x,x)\,\psi_Z(y,y)} \le \frac{n}{c_0}\phi(\delta)^2,$$

so

$$d_Z(x,y)^2 = \psi_Z(x,x) - 2\psi_Z(x,y) + \psi_Z(y,y) \le 4 \sup_{x\in A} \psi_Z(x,x) \le \frac{4n}{c_0}\,\phi(\delta)^2,$$

which shows (3.5). □

3.3 Borell–TIS and the expected supremum of Z

Now, we bound the supremum of Z over A. The tail is furnished by the Borell–TIS inequality; the expected supremum $\mathbb{E}\sup_A Z$ it requires is estimated from the finite-dimensional representation of Z (Lemma 3.5), an estimate uniform over sets A of arbitrary size.

Lemma 3.4 (Borell–TIS inequality)**.** Let $(Z(t))_{t\in A}$ be a centered Gaussian process with almost surely continuous paths on the compact set A and $\sigma_Z^2 = \sup_{t\in A} \operatorname{Var}(Z(t)) < \infty$. Then $\mathbb{E}\sup_{t\in A} Z(t) < \infty$ and, for every $u > 0$,

$$\mathbb{P}\Big(\sup_{t\in A} Z(t) \;\ge\; \mathbb{E}\sup_{t\in A} Z(t) + u\Big) \;\le \exp\!\left(-\frac{u^2}{2\sigma_Z^2}\right).$$

The estimate below holds uniformly over every set A, regardless of size, at a distance of at least δ from the design points. The subscript for the constant C_m presented below is related to the quantity m_δ it controls, in line with C_μ and $\tilde{C}_v$ that are defined later in this work.

Lemma 3.5 (Expected supremum of Z)**.** Let Z be a centered Gaussian process with covariance (3.2) and let $A \subseteq T$ be *any* set with $\operatorname{dist}\big(A, \{t_i\}_{i=1}^n\big) \geq \delta$. Under (A1)–(A2) and the hypotheses of Lemma 3.1,

$$m_\delta \;:=\; \mathbb{E} \sup_{t\in A} Z(t) \leq \sqrt{n}\,\sigma_Z \;\leq C_m\, n\, \phi(\delta), \qquad C_m := \frac{1}{\sqrt{c_0}}, \tag{3.7}$$

with C_m depending only on ϕ, n and ρ. In particular $m_\delta \to 0$ as $\delta \to \infty$, uniformly over all such A.

Proof. Write $a(t) = \phi(t_{1:n}, t)$, $M = \phi(t_{1:n}, t_{1:n})$, and let $\xi \sim \mathcal{N}(0, I_n)$. The process $t \mapsto \langle M^{-1/2} a(t), \xi\rangle$ is centered Gaussian with covariance $a(x)^\top M^{-1} a(y) = \psi_Z(x, y)$, so we may and do realize Z in this form. By Cauchy–Schwarz, *pathwise*,

$$\sup_{t\in A} |Z(t)| \;\leq \Big(\sup_{t\in A} \big\|M^{-1/2} a(t)\big\|_2\Big)\, \|\xi\|_2 \;=\; \sigma_Z\, \|\xi\|_2,$$

since $\|M^{-1/2} a(t)\|_2^2 = \psi_Z(t, t) = \operatorname{Var} Z(t)$ and $\sigma_Z = \sup_{t\in A} (\operatorname{Var} Z(t))^{1/2}$. Taking expectations and using $\mathbb{E}\|\xi\|_2 \leq (\mathbb{E}\|\xi\|_2^2)^{1/2} = \sqrt{n}$ together with the variance bound (3.3), $\sigma_Z \leq \sqrt{n/c_0}\, \phi(\delta)$, gives (3.7). □

3.4 The prior and posterior coupling

With the previous estimates in hand, we turn to the proof itself of the n-dependent result mentioned in Theorem 2.6. It rests on a coupling that places the prior and the posterior on a common probability space.

Lemma 3.6 (Coupling of prior and posterior)**.** Conditionally on $X_{1:n}$, let G be a centered Gaussian process with covariance s_n^2, as in Equation (2.1), and let Z be a centered Gaussian process with covariance ψ_Z of (3.2), independent of G. Then

$$f \overset{d}{=} G + Z, \qquad g \overset{d}{=} G + \mu_n,$$

where μ_n is the deterministic posterior mean (2.1). Under this coupling f and g live on the same probability space and share G.

Proof. By (2.1) and (3.2) we have $s_n^2 + \psi_Z = \phi$. Since G and Z are independent and centered, $G + Z$ is centered with covariance $s_n^2 + \psi_Z = \phi$, the law of f. Likewise $G + \mu_n$ has mean μ_n and covariance s_n^2, which is the conditional law of g given $X_{1:n}$. □

Remark 3.7. Under the conditional measure given $X_{1:n}$, the pair $(G+Z,\, G+\mu_n)$ realizes on a single probability space the prior law $\mathcal{GP}(0, \phi)$ and the posterior law $\mathcal{GP}(\mu_n, s_n^2)$. The first component is a fresh draw from the prior, independent of the data, not the data-generating path itself, which, conditionally on $X_{1:n}$, is of course distributed as the posterior. All conditional probabilities $\mathbb{P}(\,\cdot \mid X_{1:n})$ in Theorems 2.6 and 5.9 refer to this coupled pair, they compare the supremum of a prior draw with the supremum of a posterior draw sharing the common fluctuation G, which is exactly the comparison-of-suprema criterion discussed in Section 1.

Proof of Theorem 2.6. We work conditionally on a typical sample $X_{1:n}$ and use the coupling of Lemma 3.6. Inserting $\sup_A G(t)$ and then applying the triangle inequality, $|\sup_A u - \sup_A v| \leq \sup_A |u - v|$ twice (first with $u = G+Z,\ v = G$, then with $u = G,\ v = g$),

$$|M_A^f - M_A^g| = \Big|\sup_{t\in A}\big(G(t) + Z(t)\big) - \sup_{t\in A} g(t)\Big| \leq \sup_{t\in A} |Z(t)| + \sup_{t\in A} |G(t) - g(t)|. \tag{3.8}$$

Because $g = G + \mu_n$, the last term is deterministic: $\sup_{t\in A} |G(t) - g(t)| = \sup_{t\in A} |\mu_n(t)|$.

Mean term. Writing $a(t) = \phi(t_{1:n}, t)$ and $M = \phi(t_{1:n}, t_{1:n})$, by (3.6), Lemma 3.1 and the typical-sample bound $\|X_{1:n}\|_2 \le C_1\sqrt{n}$,

$$\begin{aligned}\sup_{t\in A}|\mu_n(t)| &= \sup_{t\in A}\big|a(t)^\top M^{-1}X_{1:n}\big| \le \sup_{t\in A}\|a(t)\|_2\,\|M^{-1}\|_{\mathrm{op}}\,\|X_{1:n}\|_2 \\ &\le \sqrt{n}\,\phi(\delta)\cdot\tfrac{1}{c_0}\cdot C_1\sqrt{n} = C_\mu\, n\,\phi(\delta),\end{aligned} \tag{3.9}$$

with $C_\mu = C_1/c_0$. Substituting (3.9) into (3.8) and using $\sup_A|Z| = \sup_A Z \vee \sup_A(-Z)$ together with $-Z \overset{d}{=} Z$,

$$\mathbb{P}\big(|M_A^f - M_A^g| \ge \tau\big) \le \mathbb{P}\Big(\sup_{t\in A}|Z(t)| \ge \tau - C_\mu n\phi(\delta)\Big) \le 2\,\mathbb{P}\Big(\sup_{t\in A} Z(t) \ge \tau - C_\mu n\phi(\delta)\Big). \tag{3.10}$$

Supremum term. We bound the right-hand side of (3.10) by the Borell–TIS inequality (Lemma 3.4) together with the expected-supremum estimate of Lemma 3.5. By Lemma 3.3, $\sigma_Z^2 \le \frac{n}{c_0}\phi(\delta)^2$, and by Lemma 3.5, $m_\delta \le C_m\, n\,\phi(\delta) \to 0$ as $\delta \to \infty$. Choose δ large enough that

$$C_\mu\, n\,\phi(\delta) + m_\delta \;\le \frac{\tau}{2}; \tag{3.11}$$

this is possible because both terms tend to 0. Setting $u = \tau - C_\mu n\phi(\delta) - m_\delta \ge \tau/2 > 0$ in Lemma 3.4,

$$\mathbb{P}\Big(\sup_{t\in A} Z(t) \ge \tau - C_\mu n\phi(\delta)\Big) = \mathbb{P}\Big(\sup_{t\in A} Z(t) \ge m_\delta + u\Big) \le \exp\!\left(-\frac{u^2}{2\sigma_Z^2}\right) \le \exp\!\left(-\frac{(\tau/2)^2}{2\sigma_Z^2}\right).$$

Using $\sigma_Z^2 \le \frac{n}{c_0}\phi(\delta)^2$,

$$\mathbb{P}\Big(\sup_{t\in A} Z(t) \ge \tau - C_\mu n\phi(\delta)\Big) \le \exp\!\left(-\frac{c_0\,\tau^2}{8\,n\,\phi(\delta)^2}\right).$$

Combining with (3.10),

$$\mathbb{P}\big(|M_A^f - M_A^g| \ge \tau \mid X_{1:n}\big) \le 2\exp\!\left(-\frac{c_0\,\tau^2}{8\,n\,\phi(\delta)^2}\right) = C\exp\!\left(-\frac{c\,\tau^2}{\phi(\delta)^2}\right), \tag{3.12}$$

with $C = 2$ and $c = \dfrac{c_0}{8n} = \dfrac{1 - n\phi(\rho)}{8n} > 0$. The constants depend only on ϕ, n and ρ, which proves (2.2). □

Remark 3.8. The dependence on the observed data $X_{1:n}$ enters (3.12) only through the geometric distance δ and through the typical-sample constant C_1. Since a typical sample occurs with probability at least $1 - e^{-c'n}$ (see Definition 2.4 and the comment below it) an unconditional bound of the same form, up to an additive error $e^{-c'n}$ for the atypical event, follows by integrating out $X_{1:n}$ over the typical event.

4 The certainty distance

Theorem 2.6 provides a non-asymptotic, quantitative version of the informal claim that GPR is reliable only near the data. Fix a tolerance $\varepsilon > 0$ and ask for the distance δ at which the right-hand side of (2.2) drops to ε. Solving $C\exp\big(-c\tau^2/\phi(\delta)^2\big) = \varepsilon$ gives $\phi(\delta)^2 = c\tau^2/\log(C/\varepsilon)$, and, since ϕ is strictly decreasing and hence invertible,

$$\delta \;=\; \phi^{-1}\!\left(\tau\sqrt{\frac{c}{\log(C/\varepsilon)}}\right), \qquad \varepsilon < C. \tag{4.1}$$

We call (4.1) the *certainty distance*. For A farther than δ from the data (design points), the suprema of prior and posterior agree up to τ with probability at least $1 - \varepsilon$, so the

observations are, by the comparison-of-suprema criterion, uninformative about extreme values beyond this range. It is a certificate of *uninformativeness* rather than of accuracy. It locates where the regression ceases to add anything to the prior, and is silent about how well prediction does inside that range (Section 6).

As with Theorem 2.6 itself, (4.1) is operative only when the resulting δ clears the *activation threshold* (3.11) (or (5.11) in the n-free form that we present later in Section 5).

Example 4.1 (Gaussian kernel)**.** For $\phi(r) = e^{-r^2/2}$ one has $\phi^{-1}(r) = \sqrt{-2\log r}$ for $r \in (0, 1]$, so (4.1) becomes

$$\delta = \sqrt{-2\log\Big(\tau\sqrt{\tfrac{c}{\log(C/\varepsilon)}}\Big)} = \sqrt{\log\frac{\log(C/\varepsilon)}{c\,\tau^2}},$$

valid whenever $\tau\sqrt{c/\log(C/\varepsilon)} \le 1$. The certainty distance thus grows only like $\sqrt{\log(1/\tau)}$ as the required accuracy τ decreases, reflecting the light tails of the Gaussian kernel.

5 An n-free sharpening of the constant

Theorem 5.9 is our second main result. It shows that the dependence on the number of observations in Theorem 2.6 can be removed under a single additional hypothesis, log-concavity of ϕ. A condition that is satisfied, for instance, by the Gaussian kernel.

Another hypothesis that changes in this section and should be flagged before we begin. The design is still ρ-separated in the geometric sense of Definition 2.2, but the non-degeneracy condition attached to it is no longer (D_n). Both conditions serve the same purpose, a quantitative floor on $\lambda_{\min}\big(\phi(t_{1:n}, t_{1:n})\big)$. Condition ($\mathrm{D}_n$) demands $\rho > \phi^{-1}(1/n)$, which grows with the number of observations, and it is used nowhere except to feed Lemma 3.1. In its place we impose the n-free condition (D) of Definition 5.1, stated below once Lemma 5.2 has supplied the quantity it constrains, and which involves the design only through ρ. Keeping (D_n) would let the hypotheses reintroduce the dependence on n that the theorem removes from the conclusion.

We develop the sharpening in dimension $d = 1$, as in the running example; the extension to $\mathbb{R}^d$ is recorded in section 5.4, where the constant remains free of n, at the price of a factor polynomial in δ that is absent on the line. The restriction is a feature of the sharpening alone, and it is worth saying why it appears only now. Notice that Theorem 2.6 is insensitive to dimension, since Lemma 3.1 bounds the row mass by $n\,\phi(\rho)$ using monotonicity of the profile and no geometry whatever, so (2.2) holds in $\mathbb{R}^d$ for every d exactly as stated. Removing n from the constant means replacing that count by a geometric one, and geometry is where dimension enters. Throughout, $\{t_i\}_{i=1}^n \subset \mathbb{R}$ are ρ-separated and $M = \phi(t_{1:n}, t_{1:n})$.

5.1 Two summable lattice constants

Define the maximal off-diagonal row mass of the Gram matrix and, for $p \in \{1, 2\}$, the maximal ℓ_p cross-covariance mass seen from a point at distance δ,

$$\Lambda(\rho) \coloneqq \max_{1\le i\le n} \sum_{j\ne i} \phi(\|t_i - t_j\|), \qquad S_p(\rho,\delta) \coloneqq \sup_{x:\,\mathrm{dist}(x,\{t_i\})\ge\delta} \sum_{i=1}^{n} \phi(\|t_i - x\|)^p. \tag{5.1}$$

With a slight abuse of notation we display only the arguments ρ and δ. Note that both quantities depend on the (fixed) design points $\{t_i\}_{i=1}^n$, but the bounds below depend on the design only through ρ and δ.

The row mass $\Lambda(\rho)$ is a property of the actual design, and through the number of terms in its defining sum it still depends on n. The point of Lemma 5.2 below is to

remove that dependence by bounding $\Lambda(\rho)$ by the single n-free lattice sum

$$\bar{\Lambda}(\rho) \;\coloneqq\; 2\sum_{k\geq 1}\phi(k\rho), \tag{5.2}$$

the value it would take on the tightest configuration ρ-separation still allows. It is this majorant, and not Λ itself, that the non-degeneracy hypothesis of the present section constrains.

Definition 5.1 (Non-degeneracy condition, n-free form)**.** A ρ-separated design $\{t_i\}_{i=1}^n \subset \mathbb{R}$ satisfies the *n-free non-degeneracy condition* if

$$\text{(D)} \qquad \bar{\Lambda}(\rho) = 2\textstyle\sum_{k\geq 1}\phi(k\rho) < 1.$$

This is the non-degeneracy hypothesis in force throughout Section 5, in place of (D_n) of Definition 2.3. Unlike (D_n), condition (D) involves the design only through ρ. By (A2) the map $\rho \mapsto \bar{\Lambda}(\rho)$ is nonincreasing and tends to 0, so (D) holds for all ρ large enough as soon as the series converges for one value of ρ.

Lemma 5.2 (Lattice bounds in dimension one)**.** Let $\{t_i\}_{i=1}^n \subset \mathbb{R}$ be ρ-separated and let ϕ satisfy (A1)–(A2). Then, for $p \in \{1, 2\}$,

$$\Lambda(\rho) \;\leq \bar{\Lambda}(\rho), \qquad S_p(\rho,\delta) \;\leq 2\sum_{k\geq 0}\phi(\delta + k\rho)^p, \tag{5.3}$$

both bounds being independent of n. If, in addition, $\log\phi$ is concave, then

$$S_p(\rho,\delta) \;\leq 2\,\Theta_p(\rho)\,\phi(\delta)^p, \qquad \Theta_p(\rho) \coloneqq \sum_{k\geq 0}\phi(k\rho)^p. \tag{5.4}$$

Note that $\bar{\Lambda}(\rho) = 2\big(\Theta_1(\rho) - 1\big)$.

Proof. Fix an index i. Along the line the inputs other than t_i lie on the two sides of t_i. By monotonicity (A2), $\sum_{j\neq i}\phi(\|t_i - t_j\|) \leq 2\sum_{k\geq 1}\phi(k\rho)$, which is the first bound. Likewise, for x with $\operatorname{dist}(x,\{t_i\}) \geq \delta$, the inputs on each side of x are ρ-separated and at distances greater or equal to $\delta + n\rho$, for $n \geq 0$, so $\sum_i \phi(\|t_i - x\|)^p \leq 2\sum_{k\geq 0}\phi(\delta + k\rho)^p$, the second bound. Finally, if $h \coloneqq \log\phi$ is concave then, since $h(0) = 0$ by (A1), h is subadditive on $[0,\infty)$, that is, for $a, b \geq 0$, $h(a) \geq \frac{a}{a+b}h(a+b)$ and $h(b) \geq \frac{b}{a+b}h(a+b)$, hence $h(a+b) \leq h(a) + h(b)$. With $a = \delta$, $b = k\rho$ this gives $\phi(\delta + k\rho) \leq \phi(\delta)\,\phi(k\rho)$, so $\phi(\delta + k\rho)^p \leq \phi(\delta)^p\phi(k\rho)^p$ and (5.4) follows from the second bound in (5.3). □

The series in (5.3) converge whenever $\sum_k \phi(k\rho) < \infty$, which holds for any covariance with $\phi(r) = \mathcal{O}(r^{-(1+\epsilon)})$ for some $\epsilon > 0$. Decay faster than r^{-1} alone does not suffice. For instance, $\phi(r) \asymp (r\log r)^{-1}$ decays faster than r^{-1}, yet $\sum_k \phi(k\rho)$ diverges. The Gaussian kernel decays much faster and gives the explicit geometric estimates used below. Displayed for later reference, the $p = 1$ case of Lemma 5.2 reads

$$S_1(\rho,\delta) \;=\; \sup_{x:\,\operatorname{dist}(x,\{t_i\})\geq\delta}\sum_{i=1}^n \phi(\|t_i - x\|) \;\leq 2\sum_{k\geq 0}\phi(\delta + k\rho) \;\leq 2\,\Theta_1(\rho)\,\phi(\delta). \tag{5.5}$$

5.2 An n-free Gram floor and variance, mean, and metric bounds

We now feed these lattice bounds into every quantity in the proof of Theorem 2.6 that still carried a factor of n, namely the Gram floor, the variance, the posterior mean, and the canonical metric.

Lemma 5.3 (n-free lower bound for the Gram matrix)**.** Under (A1)–(A2) and ρ-separation in $\mathbb{R}$, if $\bar{\Lambda}(\rho) < 1$, that is, if (D) holds, then

$$\lambda_{\min}(M) \;\geq\; \tilde{c}_0 \coloneqq 1 - \bar{\Lambda}(\rho) \;>\; 0,$$

a bound independent of n.

Before proving the lemma, observe that for the Gaussian kernel, $\bar{\Lambda}(\rho) \leq \frac{2}{e^{\rho^2/2}-1}$, so $\tilde{c}_0 \geq 1 - \frac{2}{e^{\rho^2/2}-1} > 0$ as soon as $\rho > \sqrt{2\log 3} \approx 1.482$, which is a condition independent of n. Compare this value with $\rho > \Phi^{-1}(1/n) = \sqrt{2\log n}$ in Lemma 3.1 under assumption (D_n).

Proof. $M = \phi(t_{1:n}, t_{1:n})$ has unit diagonal by (A1). By Gershgorin's theorem, equivalently the Johnson bound [2, Theorem 3] used in Lemma 3.1, and Lemma 5.2, $\lambda_{\min}(M) \geq 1 - \Lambda(\rho) \geq 1 - \bar{\Lambda}(\rho) = \tilde{c}_0$. □

The upshot of Lemma 5.3 is that the smallest eigenvalue of the Gram matrix stays bounded away from zero by a constant depending on ρ but not on n; this is the single fact whose failure made the earlier floor $c_0 = 1 - n\phi(\rho)$ of Lemma 3.1 deteriorate as n grows.

Lemma 5.4 (n-free variance bound)**.** Under the hypotheses of Lemma 5.3, the centered process Z with covariance (3.2) satisfies

$$\sigma_Z^2 = \sup_{x \in A} \mathrm{Var}\big(Z(x)\big) \;\leq \frac{S_2(\rho, \delta)}{\tilde{c}_0}.$$

For a log-concave covariance ϕ, with $\Theta_2(\rho)$ as in (5.4),

$$\sigma_Z^2 \;\leq \tilde{C}_v\, \phi(\delta)^2, \qquad \tilde{C}_v \coloneqq \frac{2\,\Theta_2(\rho)}{\tilde{c}_0},$$

a bound independent of n.

Proof. As in Lemma 3.3, with $a(x) = \phi(t_{1:n}, x)$,

$$\psi_Z(x, x) = a(x)^\top M^{-1} a(x) \;\leq \|M^{-1}\|_{\mathrm{op}}\, \|a(x)\|_2^2.$$

By Lemma 5.3, $\|M^{-1}\|_{\mathrm{op}} = 1/\lambda_{\min}(M) \leq 1/\tilde{c}_0$, while $\|a(x)\|_2^2 = \sum_i \phi(\|t_i - x\|)^2 \leq S_2(\rho, \delta)$ for every $x \in A$. Combining these bounds gives us the first bound. The second bound is a direct consequence of log-concavity and Equation (5.4). □

Two further quantities in the proof of Theorem 2.6 still carry factors of n, namely, the mean bound (3.9), $\sup_A |\mu_n| \leq (C_1/c_0)\, n\, \phi(\delta)$, and the upper bound of the expected supremum $m_\delta = \mathbb{E} \sup_A Z$ of Lemma 3.5.

Lemma 5.5 (n-free mean and canonical-metric bounds)**.** Under the same hypotheses of Lemma 5.3, with $\log \phi$ concave, for all $x, y \in A$,

$$\sup_{x \in A} |\mu_n(x)| \;\leq \frac{2\,\Theta_1(\rho)}{\tilde{c}_0}\, \phi(\delta)\, \|X_{1:n}\|_\infty, \tag{5.6}$$

$$d_Z(x, y)^2 \;\leq \frac{4L\,\Theta_1(\rho)}{\tilde{c}_0}\, \phi(\delta)\, \|x - y\|, \tag{5.7}$$

where the constants $2\Theta_1(\rho)/\tilde{c}_0$ and $4L\,\Theta_1(\rho)/\tilde{c}_0$ depend only on ϕ and ρ, and in particular neither on n nor on A.

Remark 5.6 (What is n-free in (5.6), and what is not)**.** The two bounds of Lemma 5.5 are not n-free in the same sense, and the distinction is used repeatedly below. The right-hand side of (5.7) involves the number of observations nowhere at all. The right-hand side of (5.6), by contrast, retains the sample factor $\|X_{1:n}\|_\infty$, which is a function of the n recorded values and therefore does vary with n; what the lemma asserts is that the *multiplicative constant* in front of it is n-free, not that the bound is.

Proof. Write $a(x) = \phi(t_{1:n}, x)$, $M = \phi(t_{1:n}, t_{1:n})$ and $E \coloneqq I_n - M$, so that E has zero diagonal by (A1) and $E_{ij} = \phi(\|t_i - t_j\|)$ for $i \neq j$. Since E is symmetric, its $\ell_1 \to \ell_1$ norm equals its maximal row sum, so by Lemma 5.2

$$\|E\|_{1\to 1} = \max_{1\le i\le n} \sum_{j\neq i} \phi(\|t_i - t_j\|) = \Lambda(\rho) \le \bar{\Lambda}(\rho) < 1$$

under (D). Hence the Neumann series $M^{-1} = \sum_{k\ge 0} E^k$ converges in ℓ_1 and

$$\|M^{-1}\|_{1\to 1} \ \le \sum_{k\ge 0} \bar{\Lambda}(\rho)^k = \frac{1}{1 - \bar{\Lambda}(\rho)} = \frac{1}{\tilde{c}_0}. \tag{5.8}$$

By Hölder's inequality and (5.8), followed by the ℓ_1 lattice bound (5.5),

$$\begin{aligned} |\mu_n(x)| = \big|\langle M^{-1}a(x), X_{1:n}\rangle\big| &\le \|M^{-1}a(x)\|_1 \, \|X_{1:n}\|_\infty \le \frac{\|a(x)\|_1}{\tilde{c}_0} \, \|X_{1:n}\|_\infty \\ &\le \frac{2\,\Theta_1(\rho)}{\tilde{c}_0} \, \phi(\delta) \, \|X_{1:n}\|_\infty, \end{aligned}$$

uniformly in $x \in A$, which is (5.6). Alternatively, [4] can used to lower bound the smallest eigenvalue.

To get the second inequality, put $u \coloneqq a(x) - a(y)$. By (A3) and the reverse triangle inequality $\big|\, \|t_i - x\| - \|t_i - y\| \,\big| \le \|x - y\|$ we get $\|u\|_\infty \le L\,\|x - y\|$, while (A1) and (5.5) give $\|u\|_1 \le \|a(x)\|_1 + \|a(y)\|_1 \le 4\,\Theta_1(\rho)\,\phi(\delta)$. Combining these with $\|u\|_2^2 \le \|u\|_1 \|u\|_\infty$ and with $d_Z(x,y)^2 = u^\top M^{-1} u \le \|u\|_2^2/\lambda_{\min}(M) \le \|u\|_2^2/\tilde{c}_0$ (Lemma 5.3),

$$d_Z(x,y)^2 \ \le \frac{\|u\|_1 \, \|u\|_\infty}{\tilde{c}_0} \ \le \frac{4L\,\Theta_1(\rho)}{\tilde{c}_0} \, \phi(\delta) \, \|x - y\|,$$

which is (5.7). Both constants involve only ϕ and ρ. □

Lemma 5.7 (Sharpened expected supremum)**.** Under the hypotheses of Lemma 5.3, with $\log\phi$ concave, for every set A with $\operatorname{dist}(A, \{t_i\}_{i=1}^n) \ge \delta$,

$$m_\delta \ \le \mathbb{E} \sup_{t\in A} |Z(t)| \ \le \tilde{C}_m \sqrt{2\log(2n)} \, \phi(\delta), \qquad \tilde{C}_m \coloneqq \frac{2\,\Theta_1(\rho)}{\tilde{c}_0}, \tag{5.9}$$

with m_δ defined in Equation (3.7) and $\tilde{C}_m$ depending only on ϕ and ρ.

Proof. Let $\zeta \sim \mathcal{N}(0, M)$ and put $Z(t) = a(t)^\top M^{-1}\zeta$. By Hölder's inequality, pathwise,

$$\sup_{t\in A} |Z(t)| \ \le \Big(\sup_{t\in A} \big\|M^{-1}a(t)\big\|_1\Big) \, \|\zeta\|_\infty.$$

As in the proof of Lemma 5.5, M is strictly diagonally dominant with margin $\tilde{c}_0$, so $\|M^{-1}\|_{\infty\to\infty} \le 1/\tilde{c}_0$ (Varah [4]); M^{-1} is symmetric, so its $\ell_1 \to \ell_1$ norm coincides with this, and $\|M^{-1}a(t)\|_1 \le \|a(t)\|_1/\tilde{c}_0 \le (2\Theta_1(\rho)/\tilde{c}_0)\,\phi(\delta)$ by (5.5). Finally $\mathbb{E}\|\zeta\|_\infty = \mathbb{E}\max_{i\le n}|\zeta_i| \le \sqrt{2\log(2n)}$, the standard maximal bound applied to the $2n$ standard Gaussian variables $\pm\zeta_i$. □

Remark 5.8 (The $\sqrt{\log n}$ is necessary)**.** More data no longer weakens the estimate, it only pushes out how far from the design you have to be before the estimate applies. No bound of the form $m_\delta \le c(\phi,\rho)\,\phi(\delta)$, uniform over sets A at distance δ from the design, can hold. Take $\phi(r) = e^{-r^2/2}$, $\rho = 2\delta$, $t_i = i\rho$, and let $A = \{t_i + \delta\}$ be the set of midpoints, so that $\operatorname{dist}(A,\{t_i\}) = \delta$. Each $Z(t_i+\delta)$ is centered Gaussian with variance $\big(2+o(1)\big)\phi(\delta)^2$, and the values at midpoints i and j with $|i-j|\ge 2$ are asymptotically uncorrelated as $\delta\to\infty$; Sudakov minoration applied to every other midpoint then gives $m_\delta \ge c\,\phi(\delta)\sqrt{\log n}$ for a universal $c>0$. Thus the n-dependence of (5.9) is genuine; what is n-free is the multiplicative constant $\tilde C_m$ and, decisively, the constant $\tilde c$ in the exponent of (5.10), which m_δ never touches.

5.3 The improved bound

Everything is now in place. The theorem below reruns the proof of Theorem 2.6, substituting each n-dependent estimate by its n-free counterpart from the preceding subsections.

We work throughout under the conditional measure $\mathbb{P}(\,\cdot\mid X_{1:n})$ described in Remark 3.7, on which $f = G+Z$ and $g = G+\mu_n$ are the coupled pair produced by Lemma 3.6. Since the design is fixed and the recorded values are conditioned upon, the vector $X_{1:n}$ is a fixed element of $\mathbb{R}^n$ under this measure; consequently μ_n, $\|X_{1:n}\|_\infty$, $\sigma_Z^2 := \sup_{t\in A}\operatorname{Var}\big(Z(t)\big)$ and $m_\delta := \mathbb{E}\sup_{t\in A} Z(t)$ are deterministic quantities, and the only randomness left is that of G and Z. As before we abbreviate $a(t) = \phi(t_{1:n},t)$ and $M = \phi(t_{1:n},t_{1:n})$.

Theorem 5.9 (Main result, n-free form)**.** Let $d=1$, let ϕ satisfy (A1)–(A3) with $\log\phi$ concave, and let $\{t_i\}_{i=1}^n \subset \mathbb{R}$ satisfy $\min_{i\ne j}|t_i-t_j|\ge\rho$ with $\bar\Lambda(\rho)<1$. Let $A\subset T$ be compact with $A\cap\{t_i\}_{i=1}^n=\emptyset$. Then for every $\tau>0$ and every $\delta=\operatorname{dist}(A,\{t_i\}_{i=1}^n)$ large enough (so that the activation threshold (5.11) below is cleared),

$$\mathbb{P}\big(|M_A^f - M_A^g| \ge \tau \mid X_{1:n}\big) \;\le 2\exp\!\left(-\frac{\tilde c\,\tau^2}{\phi(\delta)^2}\right), \qquad \tilde c = \frac{1}{8\tilde C_v} = \frac{\tilde c_0}{16\,\Theta_2(\rho)}, \tag{5.10}$$

with $\tilde c_0 = 1-\bar\Lambda(\rho)$ and $\tilde C_v = 2\Theta_2(\rho)/\tilde c_0$ as in Lemmas 5.3–5.4. In particular the constant $\tilde c$ depends only on ϕ and ρ, and *not* on the number of observations n.

Proof. The probabilistic content of the proof of Theorem 2.6 is unchanged; we indicate only which of its estimates are replaced and where each replacement acts. The reduction (3.8)–(3.10) uses nothing but the coupling and the inequality $|\sup_A u - \sup_A v| \le \sup_A|u-v|$, so, writing $m := \sup_{t\in A}|\mu_n(t)|$ and granting $m<\tau$ (guaranteed by (5.11) below), it holds verbatim: $\mathbb{P}\big(|M_A^f-M_A^g|\ge\tau\mid X_{1:n}\big)\le 2\,\mathbb{P}\big(\sup_{t\in A}Z(t)\ge\tau-m\big)$.

Two estimates enter the exponent. Condition (D) and Lemma 5.3 give the Gram floor $\lambda_{\min}(M)\ge\tilde c_0 = 1-\bar\Lambda(\rho)>0$ in place of $\lambda_{\min}(M)\ge c_0 = 1-n\phi(\rho)$; feeding it together with the ℓ_2 lattice bound (5.4) into the variance computation, Lemma 5.4 yields $\sigma_Z^2\le\tilde C_v\,\phi(\delta)^2$ with $\tilde C_v = 2\Theta_2(\rho)/\tilde c_0$, in place of $\sigma_Z^2\le(n/c_0)\phi(\delta)^2$ from (3.3). Both $\tilde c_0$ and $\Theta_2(\rho)$ depend on the design only through ρ.

The remaining two estimates enter only the hypothesis. In the proof of Theorem 2.6 the bounds (3.9) on m and (3.7) on m_δ serve the single purpose of ensuring, through (3.11), that the level at which Borell–TIS is applied exceeds m_δ by a definite margin. Lemmas 5.5 and 5.7 supply $m\le\tilde C_m\|X_{1:n}\|_\infty\phi(\delta)$ and $m_\delta\le\tilde C_m\sqrt{2\log(2n)}\,\phi(\delta)$, with the common constant $\tilde C_m = 2\Theta_1(\rho)/\tilde c_0$ arising because both pass through Varah's bound $\|M^{-1}\|_{\infty\to\infty}\le 1/\tilde c_0$ and the ℓ_1 lattice bound (5.5). Hence $m+m_\delta\le\tau/2$ is implied by the *activation threshold*

$$\tilde C_m\Big(\|X_{1:n}\|_\infty + \sqrt{2\log(2n)}\Big)\,\phi(\delta) \;\le\; \frac{\tau}{2}, \tag{5.11}$$

the exact analogue of (3.11) and the hypothesis of the theorem. It is not vacuous, the bracket is a finite number once the sample is conditioned upon and does not depend on δ, while $\phi(\delta) \to 0$ by (A2), so (5.11) holds for all $\delta \geq \delta_0(\tau, \phi, \rho, n, \|X_{1:n}\|_\infty)$, with δ_0 independent of A because both lemmas are uniform over sets at distance at least δ from the design. It does depend on n, unavoidably so by Remark 5.8. Note also that Lemma 5.5 pairs $\|M^{-1}a(t)\|_1$ with $\|X_{1:n}\|_\infty$ by Hölder's inequality, rather than $\|a(t)\|_2$ with $\|X_{1:n}\|_2$ by Cauchy–Schwarz as in (3.9); the sample therefore enters as a plain multiplicative factor, which is what dispenses with Definition 2.4.

Assume now that δ clears (5.11) and set $u \coloneqq \tau - m - m_\delta \geq \tau/2 > 0$, so that $\tau - m = m_\delta + u$. In the representation $Z(t) = a(t)^\top M^{-1}\zeta$ with $\zeta \sim \mathcal{N}(0, M)$ used in Lemma 5.7, the map $t \mapsto a(t)$ is continuous by (A3) and M is invertible by Lemma 5.3, so Z has continuous paths on the compact set A and $\sigma_Z < \infty$; Lemma 3.4 therefore applies and gives

$$\begin{aligned}\mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \;&\leq 2\,\mathbb{P}\Big(\sup_{t\in A} Z(t) \geq m_\delta + u\Big) \;\leq 2\exp\!\left(-\frac{u^2}{2\sigma_Z^2}\right) \\ &\leq 2\exp\!\left(-\frac{\tau^2}{8\,\tilde{C}_v\,\phi(\delta)^2}\right),\end{aligned}$$

which is (5.10) with $\tilde{c} = \tilde{c}_0/\big(16\,\Theta_2(\rho)\big)$. Since $\tilde{c}_0$ and $\Theta_2(\rho)$ are functions of ϕ and ρ alone, the constant in the exponent is free of n; the sample and the number of observations act on the conclusion only through the activation threshold (5.11). □

The split $u = \tau/2$ in (3.11) is wasteful once δ is large, because then $\sup_A |\mu_n| + m_\delta \ll \tau$. Using this, we obtain the following sharp rate.

Corollary 5.10 (Asymptotically sub-Gaussian rate)**.** Under the hypotheses of Theorem 5.9, for every $\theta \in (0,1)$ there is a δ_θ, depending on θ, τ, ϕ, ρ, on n and on the recorded sample through $\|X_{1:n}\|_\infty$, but not on A, such that for all $\delta \geq \delta_\theta$,

$$\mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \;\leq 2\exp\!\left(-\frac{\theta^2\,\tau^2}{2\,\tilde{C}_v\,\phi(\delta)^2}\right).$$

Consequently the constant in the exponent may be taken arbitrarily close to

$$\tilde{c}^\star \coloneqq \frac{1}{2\tilde{C}_v} = \frac{\tilde{c}_0}{4\,\Theta_2(\rho)}$$

as $\delta \to \infty$, four times the uniform constant $\tilde{c}$ of (5.10), and still independent of n.

Proof. Repeat the proof of Theorem 5.9, retaining the exact value $u = \tau - m - m_\delta$ rather than bounding it below by $\tau/2$. By Lemmas 5.5 and 5.7, $m + m_\delta \leq \tilde{C}_m\big(\|X_{1:n}\|_\infty + \sqrt{2\log(2n)}\big)\phi(\delta)$, whose right-hand side tends to 0 as $\delta \to \infty$ because the bracket is a finite number once the sample is conditioned upon. Hence, for any $\theta \in (0,1)$, there is a δ_θ with the stated dependence such that $m + m_\delta \leq (1-\theta)\tau$, and therefore $u \geq \theta\tau$, for all $\delta \geq \delta_\theta$. Borell–TIS (Lemma 3.4) with $\sigma_Z^2 \leq \tilde{C}_v\phi(\delta)^2$ then gives the stated bound. As in Theorem 5.9, the sample and the number of observations act only on δ_θ and not on the exponent. □

Remark 5.11 (Optimality of the asymptotic constant)**.** In the sparse-design limit $\rho \to \infty$ one has $\bar{\Lambda}(\rho) \to 0$ and $\Theta_2(\rho) \to 1$, so $\tilde{c}^\star \to \frac{1}{4}$. This limit cannot be improved. Take $n = 2$ with $t_{1,2} = x \mp \delta$ (a ρ-separated design as soon as $2\delta \geq \rho$) and $A = \{x\}$. Then $M_A^f - M_A^g = Z(x) - \mu_2(x)$, where, conditionally on $X_{1:n}$, the mean $\mu_2(x)$ is a constant and $Z(x)$ is centered Gaussian with variance $\psi_Z(x,x) = 2\phi(\delta)^2/\big(1+\phi(2\delta)\big) \to 2\phi(\delta)^2$. A bound of the form $2\exp\big(-c\,\tau^2/\phi(\delta)^2\big)$, valid for all $\tau > 0$ uniformly over ρ-separated designs, therefore forces $c \leq \frac{1}{4}(1 + o(1))$ as $\delta \to \infty$.

The constant in (5.10) is n-free, and after Lemma 5.5 so are all deterministic constants in the activation threshold (5.11) and the number of observations survives only inside the single random factor $\|X_{1:n}\|_\infty$. To convert (5.11) into an explicit requirement on δ one must therefore say something about the size of the sample, and the estimate that follows does so; it is the one place in Section 5 where a distributional assumption on $X_{1:n}$ is used. Theorem 5.9 itself makes none, the threshold being part of its hypothesis rather than of its conclusion, and the reader who prefers to keep the sample arbitrary should read it as a statement about typical data rather than as a strengthening of the theorem. The assumption is nevertheless much cheaper than Definition 2.4 since it constrains $\|X_{1:n}\|_\infty$ rather than $\|X_{1:n}\|_2$, and it is used only to place a numerical value on a threshold, never to obtain the tail bound. Since, under the prior predictive law, each $X_{t_i} \sim \mathcal{N}(0,1)$, a union bound gives $\|X_{1:n}\|_\infty \le \sqrt{2\log(2n/\eta)}$ with probability at least $1-\eta$, and on this event (5.11) is implied by $\phi(\delta) \lesssim \tau/\sqrt{\log n}$, i.e. $\delta \gtrsim \sqrt{2\log\big(\sqrt{\log n}/\tau\big)}$ for the Gaussian kernel. For comparison, the crude mean bound (3.9) gives only $\phi(\delta) \lesssim \tau/n$, i.e. $\delta \gtrsim \sqrt{2\log(n/\tau)}$. In either case the n-dependence is confined to the *location* of the regime, never to the *rate*.

5.4 Extension to $\mathbb{R}^d$ (please check)

Assume throughout this subsection that $\phi(\|\cdot\|)$ is positive definite on $\mathbb{R}^d$. Dimension enters the argument at a single point. Every step after Lemma 5.2 uses the design only through the two masses $\Lambda(\rho)$ and $S_p(\rho,\delta)$ of (5.1). It therefore suffices to replace Lemma 5.2, whose proof used the fact that on the line a point has only two sides.

Lemma 5.12 (Lattice bounds in $\mathbb{R}^d$)**.** Let $\{t_i\}_{i=1}^n \subset \mathbb{R}^d$ be ρ-separated and let ϕ satisfy (A1)–(A2). Set $\kappa_d := 4d\,3^{d-1}$. Then, for $p \in \{1,2\}$, we have

$$\Lambda(\rho) \;\le \bar\Lambda_d(\rho) := \kappa_d \sum_{k\ge 1} (1+k)^{d-1}\phi(k\rho), \qquad S_p(\rho,\delta) \;\le \kappa_d \sum_{k\ge 0} \Big(1+\tfrac{\delta}{\rho}+k\Big)^{d-1}\phi(\delta+k\rho)^p, \tag{5.12}$$

Observe that both bounds are independent of n and both series converging whenever $\sum_{k\ge1} k^{d-1}\phi(k\rho) < \infty$, in particular whenever $\phi(r) = \mathcal{O}(r^{-(d+\epsilon)})$ for some $\epsilon > 0$. If, in addition, $\log\phi$ is concave, then

$$S_p(\rho,\delta) \;\le \kappa_d\,(1+\delta/\rho)^{d-1}\,\Theta_p^{(d)}(\rho)\,\phi(\delta)^p, \qquad \Theta_p^{(d)}(\rho) := \sum_{k\ge 0}(1+k)^{d-1}\phi(k\rho)^p. \tag{5.13}$$

For $d = 1$ one may take $\kappa_1 = 2$, and (5.12)–(5.13) reduce to (5.3)–(5.4).

Proof. We first record a packing estimate. For every $x \in \mathbb{R}^d$ and every $R \ge 0$,

$$N_\rho(x,R) := \#\big\{i : \; \|t_i - x\| \in [R, R+\rho)\big\} \;\le \kappa_d\,(1+R/\rho)^{d-1}. \tag{5.14}$$

Indeed, ρ-separation makes the balls $B(t_i,\rho/2)$ pairwise disjoint, and each of these balls indexed by the left-hand side of (5.14) is contained in the annulus $B\big(x, R+\frac{3}{2}\rho\big)\setminus B\big(x,(R-\frac{1}{2}\rho)_+\big)$. Comparing volumes and applying the mean value theorem to $t \mapsto t^d$ on an interval of length 2ρ,

$$N_\rho(x,R) \le \frac{\big(R+\frac{3}{2}\rho\big)^d - \big((R-\frac{1}{2}\rho)_+\big)^d}{(\rho/2)^d} \;\le \frac{2\rho\,d\,\big(R+\frac{3}{2}\rho\big)^{d-1}}{(\rho/2)^d} \;\le 4d\,3^{d-1}\big(1+R/\rho\big)^{d-1},$$

the last step using $R+\frac{3}{2}\rho \le \frac{3}{2}\rho\,(1+R/\rho)$.

Fix i and group the remaining inputs by the annuli of (5.14) centred at t_i with $R = k\rho$, $k \ge 1$; every t_j with $j \ne i$ falls in one of them by ρ-separation, and $\phi(\|t_i - t_j\|) \le \phi(k\rho)$

by (A2). Summing gives the first bound in (5.12). Likewise, for x with $\operatorname{dist}(x, \{t_i\}) \geq \delta$, grouping by the annuli centred at x with $R = \delta + k\rho$, $k \geq 0$, and using $\phi(\|t_i - x\|)^p \leq \phi(\delta + k\rho)^p$ gives the second. Convergence of both series follows from $(1+k)^{d-1} \leq 2^{d-1}k^{d-1}$ for $k \geq 1$ and $\phi \leq 1$.

Finally, if $\log \phi$ is concave then $\phi(\delta + k\rho) \leq \phi(\delta)\,\phi(k\rho)$, as in the proof of Lemma 5.2, while $1 + \delta/\rho + k \leq (1 + \delta/\rho)(1 + k)$; substituting both into the second bound of (5.12) yields (5.13). □

Accordingly, condition (D) of Definition 5.1 is replaced in $\mathbb{R}^d$ by the n-free inequality

$$(\mathrm{D}^d) \qquad \bar{\Lambda}_d(\rho) < 1,$$

under which Lemma 5.3 gives the Gram floor $\lambda_{\min}(M) \geq \tilde{c}_0^{(d)} \coloneqq 1 - \bar{\Lambda}_d(\rho) > 0$. Condition ($\mathrm{D}^d$) is still free of n, but it demands a separation growing with the dimension. For the Gaussian kernel it holds as soon as $\rho \gtrsim \sqrt{2 \log \kappa_d} = \mathcal{O}(\sqrt{d})$.

Theorem 5.13 (n-free bound in $\mathbb{R}^d$)**.** Let ϕ satisfy (A1)–(A3) with $\log \phi$ concave and $\phi(r) = \mathcal{O}(r^{-(d+\epsilon)})$, let $\phi(\|\cdot\|)$ be positive definite on $\mathbb{R}^d$, and let $\{t_i\}_{i=1}^n \subset \mathbb{R}^d$ be ρ-separated and satisfy (D^d). Let $A \subset T$ be compact with $A \cap \{t_i\}_{i=1}^n = \emptyset$ and put $\delta = \operatorname{dist}\big(A, \{t_i\}_{i=1}^n\big)$. Then, with the effective scale

$$\Psi_d(\delta) \coloneqq (1 + \delta/\rho)^{(d-1)/2}\,\phi(\delta),$$

for every $\tau > 0$ and every big enough $\delta > 0$,

$$\mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \;\leq 2\exp\!\left(-\frac{\tilde{c}_d\,\tau^2}{\Psi_d(\delta)^2}\right), \qquad \tilde{c}_d = \frac{\tilde{c}_0^{(d)}}{8\,\kappa_d\,\Theta_2^{(d)}(\rho)}, \tag{5.15}$$

with $\tilde{c}_d$ depending only on ϕ, ρ and d, and in particular neither on the number of observations n nor on A. For $d = 1$, $\Psi_1 = \phi$ and (5.15) is (5.10).

Proof. The proof of Theorem 5.9 applies verbatim once each appeal to Lemma 5.2 is replaced by the corresponding bound of Lemma 5.12; we record only what the substitution changes. The exponent comes from the variance alone. By Lemma 5.4 and (5.13),

$$\sigma_Z^2 \;\leq \frac{S_2(\rho, \delta)}{\tilde{c}_0^{(d)}} \;\leq \tilde{C}_v^{(d)}\,\Psi_d(\delta)^2, \qquad \tilde{C}_v^{(d)} \coloneqq \frac{\kappa_d\,\Theta_2^{(d)}(\rho)}{\tilde{c}_0^{(d)}},$$

and $\tilde{c}_d = 1/\big(8\tilde{C}_v^{(d)}\big)$. The mean and expected-supremum bounds of Lemmas 5.5 and 5.7 pass instead through the ℓ_1 mass S_1, hence through the full factor $(1 + \delta/\rho)^{d-1}$. Thus, writing $\tilde{C}_m^{(d)} \coloneqq \kappa_d\,\Theta_1^{(d)}(\rho)/\tilde{c}_0^{(d)}$, the threshold (5.11) becomes

$$\tilde{C}_m^{(d)}\Big(\|X_{1:n}\|_\infty + \sqrt{2\log(2n)}\Big)(1 + \delta/\rho)^{d-1}\phi(\delta) \;\leq \frac{\tau}{2}. \tag{5.16}$$

Convergence of $\sum_k k^{d-1}\phi(k\rho)$ forces its summand to vanish, and ϕ is nonincreasing, so $\delta^{d-1}\phi(\delta) \to 0$ as $\delta \to \infty$, consequently (5.16) holds for all δ large enough, uniformly over sets A at distance at least δ from the design. Since the threshold enters the hypothesis and not the exponent, the constant $\tilde{c}_d$ is unaffected by n. □

6 Interpretation: influence regions and the certainty distance

Conditioning has a *local effect*. Near the observed inputs the posterior $g \sim \mathcal{GP}(\mu_n, s_n^2)$ is pinned to the data and its variance is suppressed; far from them the posterior mean reverts to the prior mean and the posterior variance returns to the prior variance

$\phi(0) = 1$. This pointwise reversion is classical, and it is what Figure 1 displays. The posterior paths hug the data and fan back out to the prior envelope as the distance to the observations grows. Theorems 2.6 and 5.9 make this familiar marginal statement quantitative at the level of the *supremum*, that is, the extreme value attained over a whole region A forgets the data at the rate $\phi(\delta)$, where $\delta = \operatorname{dist}(A, \{t_i\}_{i=1}^n)$.

6.1 The influence region

It is convenient to read either main theorem as delimiting a region of the index set within which the observations still carry information about extremes. Both have the form $\mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \leq C \exp\big(-c\,\tau^2/\phi(\delta)^2\big)$, and the definition below depends on them only through the pair (C, c).

Definition 6.1 (Influence region and influence radius)**.** Fix an accuracy $\tau > 0$ and a confidence level $1 - \varepsilon$. The *influence radius* is the certainty distance (4.1),

$$\delta_\star(\tau, \varepsilon) \;=\; \phi^{-1}\!\left(\tau \sqrt{\frac{c}{\log(C/\varepsilon)}}\right), \qquad \varepsilon < C,$$

and the *influence region* is its neighborhood of the data,

$$\mathcal{I}_\star(\tau, \varepsilon) = \big\{x \in T : \operatorname{dist}(x, \{t_i\}_{i=1}^n) \leq \delta_\star(\tau, \varepsilon)\big\}.$$

Here $(C, c) = \big(2,\, c_0/(8n)\big)$ under the hypotheses of Theorem 2.6, and $(C, c) = (2, \tilde{c})$ with $\tilde{c} = \tilde{c}_0/\big(16\,\Theta_2(\rho)\big)$ under those of Theorem 5.9; in the latter case $\tilde{c}$ may be replaced by the larger $\tilde{c}^\star = \tilde{c}_0/\big(4\,\Theta_2(\rho)\big)$ of Corollary 5.10 at the price of a further restriction on δ.

Informally, $\mathcal{I}_\star$ is the tube of radius $\delta_\star$ around the design points, the region within which the observations may still relate to the extremes. By (2.2), or by (5.10) in the sharpened form, every compact A lying *outside* $\mathcal{I}_\star$ inherits a guarantee that its prior and posterior suprema coincide up to τ:

$$\operatorname{dist}\big(A, \{t_i\}_{i=1}^n\big) \;\geq\; \delta_\star(\tau, \varepsilon) \qquad \Longrightarrow \qquad \mathbb{P}\big(|M_A^f - M_A^g| \geq \tau \mid X_{1:n}\big) \;\leq \varepsilon, \tag{6.1}$$

provided δ also clears the activation threshold (3.11), respectively (5.11), of the theorem invoked. The influence radius is therefore an explicit distance beyond which the regression is, by the comparison-of-suprema criterion, indistinguishable from sampling the prior. The implication in (6.1) is a certificate of *uninformativeness* rather than of accuracy. Outside $\mathcal{I}_\star$ the observations are, by this criterion, uninformative about the extremes, whereas inside $\mathcal{I}_\star$ the bound is silent, as it should be, that being precisely the regime in which the observations are meant to matter.

6.2 How the radius scales

The influence radius behaves as intuition demands. It *grows* when the accuracy τ is tightened or the confidence $1-\varepsilon$ is raised, and it *shrinks* as the prior correlation decays faster, since a short-range kernel forgets sooner. For the Gaussian kernel $\phi(r) = e^{-r^2/2}$ the Example of Section 4 gives the closed form

$$\delta_\star(\tau, \varepsilon) = \sqrt{\log \frac{\log(C/\varepsilon)}{c\,\tau^2}} \;=\; \mathcal{O}\big(\sqrt{\log(1/\tau)}\big),$$

so halving the tolerated discrepancy τ pushes the frontier outward by only $\mathcal{O}(1/\delta_\star)$. The influence region has a sharp, almost fixed horizon, a direct consequence of the light tails of ϕ. This scaling law is insensitive to the constant, since $\delta_\star$ enters only through ϕ^{-1}; what the two theorems disagree about is the numerical location of the horizon.

In the form given by Theorem 2.6 that location degrades with the sample, the constant $c = c_0/(8n)$ depending on n both directly and through the requirement $\rho > \phi^{-1}(1/n)$ of condition (D_n). Theorem 5.9 removes this. Under (D) and log-concavity the constant $\tilde{c}$ depends only on ϕ and ρ, so the scale constant is *independent of the number of observations*. Adding more ρ-separated data does not extend the reach of the model, it only fills in the interior of $\mathcal{I}_\star$.

Three limitations are worth stating plainly. First, the bound is *one-sided*, it certifies when GPR adds nothing beyond the prior, not when it predicts well. Second, it rests on ρ-separation, in the general form of Theorem 2.6 together with the non-degeneracy condition (D_n) and a typical sample (Definition 2.4), and in the sharpened form of Theorem 5.9 together with the n-free condition (D) alone, Lemma 5.5 having removed the typicality assumption; in either case heavily clustered or near-coincident inputs lie outside the present analysis. Third, the radius is measured in the prior's own metric, through ϕ^{-1}, so a misspecified kernel, in particular one whose tail is heavier than the truth, will report an influence region larger than the data genuinely warrants.

Acknowledgments. The first author thanks Sayan Mukherjee for being the catalyst behind this problem.

References

[1] Adler, R. J. and Taylor, J. E.: *Random Fields and Geometry*. Springer Monographs in Mathematics, Springer, New York, 2007.

[2] Johnson, C. R.: A Geršgorin-type lower bound for the smallest singular value. *Linear Algebra Appl.* **112**, (1989), 1–7.

[3] Tanguy, K.: Some superconcentration inequalities for extrema of stationary Gaussian processes. *Statist. Probab. Lett.* **106**, (2015), 239–246. arXiv:1503.06563

[4] Varah, J. M.: A lower bound for the smallest singular value of a matrix. *Linear Algebra Appl.* **11**, (1975), 3–5.

[5] Datta, A., Banerjee, S., Finley, A. O. and Gelfand, A. E.: Hierarchical nearest-neighbor Gaussian process models for large geostatistical datasets. *J. Amer. Statist. Assoc.* **111**, (2016), 800–812.

[6] Stein, M. L.: *Interpolation of Spatial Data: Some Theory for Kriging*. Springer Series in Statistics, Springer, New York, 1999.

[7] Stein, M. L.: The screening effect in kriging. *Ann. Statist.* **30**, (2002), 298–323.

[8] Stein, M. L.: 2010 Rietz Lecture: When does the screening effect hold? *Ann. Statist.* **39**, (2011), 2795-2819 .

[9] Vecchia, A. V.: Estimation and model identification for continuous spatial processes. *J. Roy. Statist. Soc. Ser. B* **50**, (1988), 297–312.

[10] Chatterjee, S.: *Superconcentration and Related Topics*. Springer Monographs in Mathematics, Springer, Cham, 2014.

[11] Chernozhukov, V., Chetverikov, D. and Kato, K.: Comparison and anti-concentration bounds for maxima of Gaussian random vectors. *Probab. Theory Related Fields* **162**, (2015), 47–70.

[12] Giessing, A.: Anti-concentration of suprema of Gaussian processes and Gaussian order statistics. Preprint, (2023). arXiv:2310.12119v1

[13] Lederer, A., Umlauft, J. and Hirche, S.: Uniform error bounds for Gaussian process regression with application to safe control. Preprint, (2019). arXiv:1906.01376

[14] Lederer, A., Umlauft, J. and Hirche, S.: Posterior variance analysis of Gaussian processes with application to average learning curves. Preprint, (2019). arXiv:1906.01404

[15] Liu, J. and Kok, S.: Practical global and local bounds in Gaussian process regression via chaining. In: *The Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)*. arXiv:2511.09144

[16] van der Vaart, A. W. and van Zanten, J. H.: Rates of contraction of posterior distributions based on Gaussian process priors. *Ann. Statist.* **36**, (2008), 1435–1463.

[17] Coles, S.: *An Introduction to Statistical Modeling of Extreme Values*. Springer Series in Statistics, Springer, London, 2001.

[18] de Haan, L. and Ferreira, A.: *Extreme Value Theory: An Introduction*. Springer Series in Operations Research and Financial Engineering, Springer, New York, 2006.

[19] Bull, A. D.: Convergence rates of efficient global optimization algorithms. *J. Mach. Learn. Res.* **12**, (2011), 2879–2904. arXiv:1101.3501

[20] Kimeldorf, G. S. and Wahba, G.: A correspondence between Bayesian estimation on stochastic processes and smoothing by splines. *Ann. Math. Statist.* **41**, (1970), 495–502.